

Dhruv Nirmal

Melbourne, Australia | dhruvnirmal2111@gmail.com | +61 406 259 619 | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

I build and validate forecasting and predictive models that a business actually uses to make decisions, and I dig into the underlying factors until I can explain why a model moves, not just what it predicts. I built a Prophet demand-forecasting model with external regressors that held within a 12.5 to 14% error margin against actuals, and I led a logistic-regression classifier over roughly **A\$12B** of client spend; both started as a vague question, moved through SQL and Python prototyping and validation on held-out data, and ended in a recommendation senior stakeholders acted on. I also design and own MyFacit, a live subscription-style product for independent cafes, where I define the unit economics it tracks (revenue, cost, margin) and turn a weekly number into the one decision an owner should make next. That is the closest experience I have to owning a metric end to end for a product, and it is the same discipline behind owning and evolving StellarTech's LTV prediction algorithm: build a forecast, validate it against reality, and work out what is actually driving it.

KEY SKILLS

Forecasting & Predictive Modelling: Prophet time-series forecasting, external-regressor and feature engineering, backtesting against held-out data, error-margin validation, factor analysis

Statistics & Model Validation: classification modelling (logistic regression, TF-IDF and embeddings), GridSearchCV, cross-validation, class-imbalance handling, precision/recall/F1, hypothesis-driven analysis

SQL & Data: SQL (complex queries, prototyping at scale, large datasets), Python (pandas, numpy, scikit-learn), Snowflake, AWS S3

Product & Unit Economics: unit economics (revenue, cost, margin, payback thinking), owning product metrics end to end, pricing and cost analysis, menu engineering

Visualisation & Stakeholder Communication: Tableau, Power BI, dashboard design, presenting to technical and senior business stakeholders

EXPERIENCE

Data Analyst, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 to Present | Melbourne, Australia

A data and analytics consulting firm serving enterprise clients across Australia and New Zealand. I own analytical engagements end to end: framing the question, building and validating the model, and presenting the finding to business stakeholders. Selected engagements:

Demand forecasting and model validation, fresh produce and agricultural client

- Problem: the client needed a rolling three-month view of raw-material and chemical inventory to plan ordering, including items sourced internationally and exposed to freight-cost swings.
- Approach: built a Prophet time-series forecasting model, engineered external regressor features (sea-freight trends, input pricing) in Python, prototyped the underlying query logic in SQL, and backtested the forecast against actual usage on held-out periods before handing it to a planning dashboard.
- Result: held within a 12.5 to 14% error margin against actuals, comfortably inside the client's tolerance; the client used it to plan ordering three months ahead.

ML classification and factor analysis, enterprise procurement client (~A\$12B spend)

- Problem: only 60 to 65% of a five-year, roughly A\$12B spend book was reliably categorised, and the manual process took an account manager over a month per cycle.

- Approach: led the company's first ML classifier for this problem, a one-vs-all logistic regression per category built in scikit-learn. Engineered features on top of TF-IDF and embeddings text signals (spend value, line count) to isolate what actually distinguished a category, tuned via GridSearchCV and cross-validation, and explicitly handled the class imbalance across categories.
- Validation/Result: evaluated on precision, recall and F1 given the imbalance rather than raw accuracy; cut the cycle to a single day's model run and documented the pipeline, now rolled out across the wider client base.

Pricing and procurement analytics, multi-venue restaurant group (AU/NZ)

- Problem: the client felt supply costs were too high but had no product-level view of where the money was leaking.
- Approach: owned the account solo as the sole analyst. Structured a product catalog of roughly 3,000 items in SQL, built a pricing-decision dashboard tracking price movement and cost leakage by product, venue and supplier, and ran the client relationship directly through bi-weekly and monthly stakeholder meetings.
- Result: presented findings to the Chief Procurement Officer and category managers; helped the client identify close to 30% savings, roughly **A\$2M**, in their largest category over a year.

Also own the end-to-end SQL pipeline and data-validation logic for the firm's largest client account (roughly 6M invoice rows refreshed weekly across 5 global regions), including automated checks that catch discrepancies before they reach a report and coordination with the firm's own data engineering team on the underlying refresh logic.

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 to Nov 2023 | Melbourne, Australia

- Built a distributed anomaly-detection pipeline (Databricks, PySpark) on government procurement data, improving detection accuracy by about 20%, and delivered the findings through a Power BI solution adopted by senior Department of Transport stakeholders.

Research Data Analyst, Terminal Ballistics Research Laboratory (TBRL), DRDO

Jan 2021 to Jul 2021

- Built statistical models to predict blast-wave noise levels, improving prediction accuracy by about 20%, and compiled findings into a structured report used to refine defence equipment design.

PROJECTS

MyFacit: a multi-tenant profitability product I built and own, for independent cafes

- Designed and shipped the product end to end: defined the unit economics it tracks (revenue, wage cost %, food cost %, gross margin) and built the data model that unifies point-of-sale, wages and 14 suppliers' invoices into one weekly view.
- Turned that data into a weekly decision the owner acts on: a "pulse" combining revenue forecasting, menu engineering (which items actually carry the margin), supplier price-creep detection, and one recommended action for the week ahead, rather than a static report.
- In active use with a design-partner cafe (Neighbours Cafe, St Kilda), iterating the product from real operating data through weekly reviews. My closest analog to owning the prediction logic and unit economics for a live, subscription-style product.

EDUCATION

Master of Data Science, Monash University | Feb 2022 to Dec 2023

Coursework: Statistics I/II, Machine Learning, Applied Forecasting, Communicating with Data, High Dimensional Analysis.

Bachelor of Engineering, Thapar University | May 2017 to May 2021